

JTMK Project Report:

Smart Attendance Monitoring System Using Face Recognition

Project Title:

Smart Attendance Monitoring System with Real-Time Face Recognition

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Introduction:

This project presents a smart face recognition-based attendance monitoring system designed to enhance classroom management at JTMK. The system uses a camera and machine learning to automatically detect and record student attendance in real time, eliminating manual roll calls.

Objectives:

1. To develop a face recognition system using OpenCV and Python.
2. To automate the student attendance process using a webcam.
3. To store and manage attendance data securely in a database.
4. To improve time efficiency and reduce proxy attendance.

Tools and Technologies Used:

- Hardware: Webcam, Laptop
- Software: Python, OpenCV, SQLite, Tkinter GUI
- Programming Language: Python

Features:

- Real-time face detection and recognition
- Secure student database with photos and ID
- Attendance timestamp logging
- GUI for manual review and correction

Implementation Process:

1. Captured student facial data and trained the recognition model.
2. Developed Python script using OpenCV for face matching.
3. Designed a simple GUI to show attendance logs.
4. Integrated SQLite to store records.
5. Tested system accuracy with different lighting and angles.

Results:

- Attendance captured with 94% accuracy.
- System response time: ~2 seconds per student.
- Successfully identified and marked 30 students per session.

Challenges:

- Poor lighting affected recognition accuracy.
- Training model required multiple image samples.
- Real-time performance optimization.

Future Enhancements:

- Integrate with cloud storage and analytics dashboard.
- Add voice notification for marked attendance.
- Mobile app support for remote monitoring.

Conclusion:

The system offers an efficient and modern approach to classroom attendance, helping educators in JTMK save time while maintaining reliable records.

References:

1. OpenCV Documentation – <https://docs.opencv.org>
2. Python SQLite3 – <https://docs.python.org/3/library/sqlite3.html>
3. Face Recognition with Python – <https://realpython.com/face-recognition-with-python/>